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A study of constituent redistribution in U–Zr fuels using quantitative phase-field modeling and sensitivity analysis

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HIGHLIGHTS

- Constituent redistribution models must account for swelling and sodium infiltration.
- Swelling and sodium infiltration influence Zr bathtub size, shape, and location.
- Phase kinetic parameters scale the magnitude of constituent redistribution.
- Constituent redistribution profiles are most sensitive to fuel surface temperature.

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ABSTRACT

Constituent redistribution in U–Zr fuels produces radially-distributed phase fields, each with different material properties and behaviors. The location and composition of the phase fields evolve dynamically due to the influence of swelling, fission gas release, and sodium infiltration on the thermal conductivity of the fuel. Gaps in the understanding of these processes limit the ability to model higher-level behaviors such as fuel-cladding chemical interaction and perform design and safety analyses with confidence. In the current work, we developed a quantitative phase-field model of macroscale constituent redistribution in the U–Zr system. Model parameters were optimized, and the model was validated against an independent dataset. No modification of the phase transition temperatures was necessary, but diffusion needed to be enhanced to reproduce the observed behaviors. Sensitivity analysis was then used to investigate gaps in the understanding of the system and identify the most impactful parameters and behaviors. The mobile interface between the γ and $\beta+\gamma$ phase fields was found to be particularly influential in forming the Zr bathtub. The β and γ kinetic parameters scale the overall magnitude of constituent redistribution, while parameters associated with swelling and sodium infiltration influence the size, shape, and location of the Zr bathtub. These findings were used to justify specific recommendations for improving U–Zr modeling capability, which will expedite fuel development and qualification.
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1. Introduction

Most of the current knowledge of metallic nuclear fuels was gathered during Argonne National Laboratory's Integral Fast Reactor (IFR) program in the 1980s and 1990s [1–4]. Substantial improvements in metallic fuel performance and safety were achieved by applying engineering solutions developed from IFR experimental data and modeling studies. These efforts also laid the foundation for early metallic fuel performance codes [5–8].

The material properties and irradiation performance of metallic fuels such as U–Zr and U-Pu-Zr are strongly influenced by the complex behaviors of the crystalline phases. This is in contrast to oxide fuels, the properties and performance of which are dominated by the grain structures. The phases present in a metallic fuel element and the ratios between them vary dynamically throughout the fuel's life. They respond to (and influence) swelling, fission gas release, sodium infiltration, and constituent redistribution. Numerous studies have been conducted to investigate the physics underlying these behaviors and to characterize their effects [1–4,9].

Swelling, fission gas release, and sodium infiltration affect the fuel's thermal conductivity by tens of percent. This alters the

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temperature gradient across the fuel, and the temperature gradient promotes formation of radially-distributed phase fields. The fuel's constituents redistribute in response to the temperature gradient and chemical potential gradients, further affecting its thermal conductivity [10]. Constituent redistribution has been the subject of various studies dating back to the IFR program [11–16], and their findings and models are still used by modern metallic fuel performance codes today [17,18].

The complexity of metallic fuel behaviors has forced researchers to make broad assumptions, rely heavily on empirical correlations, and use numerous modeling parameters to improve the accuracy of constituent redistribution models. These features limit the ability to investigate unexplained behaviors and model higher-level behaviors, including fuel-cladding chemical interaction. Recently, phase-dependent material property data and advanced modeling techniques such as the quantitative phase-field method have been applied to help refine the understandings of these systems [19–21]. These findings will allow extrapolation beyond the existing experimental data for the design and safety analyses that will be required to support development of new sodium fast reactors [22–24].

In the present work, we leverage what has been learned from previous constituent redistribution studies and apply new modeling techniques to develop a quantitative phase-field model for constituent redistribution in U–Zr. We incorporate all of the most up to date material property data to refine or replace many of the assumptions, empirical correlations, and modeling parameters used in previous studies. The resulting model is used to predict the dynamic coevolution of phases, constituents, and temperature throughout the fuel, providing additional insight into the in-core behavior of U–Zr fuels.

Optimization techniques are applied to account for parameter uncertainties and/or proposed irradiation effects, and the model is then validated against an independent experimental dataset. Similarities and discrepancies are used to evaluate the validity of common assumptions and simplifications and to identify the most influential physical processes and material properties. Finally, sensitivity analysis is conducted to quantify these findings and to justify recommendations for future modeling and experimental efforts. These recommendations will increase the effectiveness of data collection, iteratively improving the understanding of the U–Zr system and ability to accurately model its behavior.

2. Model development

Previous constituent redistribution studies utilized parameters such as interdiffusion coefficients, heats of transport, and simplified phase diagrams and/or activity coefficients to model the thermochemical forces that drive constituent redistribution. Modifications to the kinetic and thermodynamic parameters were necessary to make predictions consistent with Post-Irradiation Examinations (PIEs) of the metallic fuel elements. Kinetic quantities were used as modeling parameters or optimized to account for deficiencies in the available experimental data, parameter uncertainties, and/or the effects of proposed irradiation-enhanced diffusion. Some studies allowed phase transition temperatures to vary by as much as 10% to account for parameter uncertainties and/or irradiation effects.

The current work closely follows that of Hofman et al. [12], whose 1996 constituent redistribution model was the only one developed exclusively for U–Zr fuels. Hofman et al. examined PIE data from two U–10Zr wt% fuel elements, which were irradiated in Experimental Breeder Reactor II during Experiment X447: DP-81 and DP-11 [25]. We use the same two datasets to optimize and validate our model and take advantage of burnup, power, fuel surface temperature, and fuel geometry values given by Hofman et al.

Typical U–Zr fuel alloys consist of about 10 wt% (about 22 at%) Zr and are subjected to temperatures of about 620–750°C under normal operating conditions. This portion of the U–Zr system is shown by the partial phase diagram included in Fig. 1, which was calculated from a recent thermodynamic assessment [26]. The diagram shows that three phases are expected to influence constituent redistribution in U–Zr. These are orthorhombic α, tetragonal β, and body-centered cubic γ. γ has been shown to decompose into U-rich γ_1 and Zr-rich γ_2 above 693°C [27], but the existence of this miscibility gap has been neglected in all but one of the previous constituent redistribution studies [11]. Its omission from the system's thermodynamic description is not expected to affect the model's predictions because no γ_1 has been observed in PIEs of U–Zr fuel elements [12]. As such, the miscibility gap is not shown in Fig. 1, and the current work assumes that γ has unlimited solubility for both U and Zr.

As noted in Sec. 1, previous studies have demonstrated that constituents redistribute in response to spatial gradients in both temperature and chemical potential. In the current work, quantitative phase-field techniques are used to mechanistically predict the coevolution of phases and constituents within simulated fuel elements subjected to dynamic heat generation and transport. The relevant physical relationships and material-specific properties are introduced in Subsecs. 2.1 and 2.2.

2.1. Phase-field components and parameters

The phase-field method employs diffuse interfaces to spatially resolve the structural features that influence the behavior of a material [28]. Use of diffuse interfaces eliminates explicit interface tracking and offers reduced computational complexity when compared to sharp interface models. It also makes it easier to directly employ thermodynamic and kinetic data describing the system. Use of a diffuse interface width sufficiently smaller than the domain size ensures that phase-field model predictions are consistent with those obtained using sharp interface models.

Material-specific thermodynamic and kinetic properties can be incorporated into phase-field models to make quantitative predictions about the evolution of real systems. Drawbacks of this

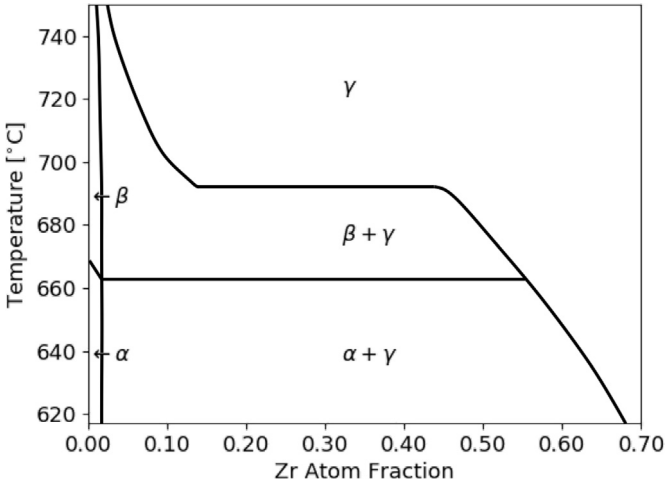


Fig. 1. Partial U–Zr phase diagram calculated from a thermodynamic assessment [26]. The γ_1-γ_2 miscibility gap has been omitted because no γ_1 has been observed during PIEs of U–Zr fuel elements [12]. The phase diagram shows that three phases are expected to influence constituent redistribution in typical U–Zr fuel alloys under normal operating conditions: α, β, and γ.

approach include the need to define additional model parameters and use of higher-order differential equations, which increases computational expense over traditional macroscale thermo-chemical diffusion models. Several in-depth reviews of these modeling techniques are available [29–32].

The structural features of interest for modeling constituent redistribution are the phase and constituent compositions throughout the fuel. Model parameters were therefore chosen to resolve only the interfaces between the different single- and multi-phase regions. Individual grains within each region are not resolved separately in the current work, allowing the entire radius of a fuel slug to be modeled.

In the phase-field method, the atom fractions of alloy constituents are represented by conserved variable fields. The independent variable c was chosen to represent the atom fraction of Zr. The two constituent atom fractions must sum to one, and the dependent atom fraction of U can therefore be obtained by $1-c$. The evolution of c is governed by the modified Cahn-Hilliard equation, which accounts for Zr migration due to spatial gradients in both chemical potential and temperature [33,34]:

$$\frac{\partial c}{\partial t} = \nabla \cdot \left(M_c \nabla \frac{\delta F}{\delta c} - M_T \frac{\nabla T}{T} \right), \tag{1}$$

where t is the time, M_c is the chemical mobility, F is the free energy functional, M_T is the thermal mobility, and T is the temperature.

Phase fractions are represented by non-conserved variable fields. γ was chosen to represent the phase fraction of γ, and its evolution is governed by the Allen-Cahn equation:

$$\frac{\partial \gamma}{\partial t} = -L \frac{\delta F}{\delta \gamma}, \tag{2}$$

where L is the phase mobility. L is used as a modeling parameter in the current work because there are no data available to define it. A value of $L = 2.5 \times 10^{-10}\ \mathrm{m^3/(J\ s)}$ was chosen by using sensitivity analysis techniques to minimize the model's overall sensitivity to L, yielding diffusion-controlled processes.

Normally, a three-phase model would require three additional non-conserved variables: two to represent the phase fractions of α and β and a third to constrain the sum of the three phase fractions to one. The system is simplified considerably by letting γ be the independent phase fraction and defining the sum of the α and β phase fractions as $\alpha+\beta = 1-\gamma$. A switching function ($h_{\alpha\beta}$) is then used to smoothly interpolate between α and β at the transition temperature between the $\alpha+\gamma$ and $\beta+\gamma$ phase fields shown in Fig. 1 $h_{\alpha\beta}$ is defined as follows:

$$h_{\alpha\beta} = 1.0 - \frac{1}{2}\tanh\left(\frac{T - T_{\alpha\beta}}{2}\right), \tag{3}$$

where $T_{\alpha\beta} = 662^{\circ}\mathrm{C}$ is the transition temperature between the two phase fields. The dependent phase fractions can then be obtained by $\alpha = (1-h_{\alpha\beta})(1-\gamma)$ and $\beta = h_{\alpha\beta}(1-\gamma)$. This simplification decreases the size of the nonlinear system by almost half, greatly reducing the computational expense. Its use does not negatively impact the model's predictions because the equilibrium atom fractions of α and β are nearly equal, and the two phases coexist only within a very narrow range of compositions and temperatures.

The free energy functional describes the thermodynamic behavior of the system in terms of its total free energy, which includes both bulk and interfacial contributions. It is further defined as follows:

$$F = \int_V \left(f_{bulk} + f_{grad} \right) dV, \tag{4}$$

where V is the system volume, f_{bulk} is the bulk free energy, and f_{grad} is the gradient free energy. Equations (1) and (2) require the total free energy of the system to decrease with time, ensuring that the constituent atom fractions and phase fractions evolve toward thermodynamic equilibrium.

The change in the free energy due to a change in the amount of a constituent present is the chemical potential. Spatial gradients in the chemical potential drive constituents to migrate at a rate defined by the chemical mobility. This takes the place of the Fickian contributions modeled in previous studies. The bulk free energy is given by:

$$f_{bulk} = \frac{1}{V_m^{ref}} \left[(1-h) f_{\alpha\beta} + h f_{\gamma} \right] + wg, \tag{5}$$

where $V_m^{ref} = 1.249 \times 10^{-5}\ \mathrm{m^3/mol}$ is the reference molar volume, which was calculated for α-U. Here also, h is an interpolation function (not to be confused with $h_{\alpha\beta}$). $f_{\alpha\beta}$ is a bulk free energy function representing the thermodynamic behavior of α and β, and f_{γ} is a bulk free energy function representing γ. Finally, w is the barrier height, and g is a barrier function. The bulk free energy functions are derived from CALPHAD free energies [35] and are discussed in further detail at the end of this subsection.

The interpolation and barrier functions have standard forms and are further defined as follows:

$$h = \gamma^3 \left(6\gamma^2 - 15\gamma + 10 \right), \tag{6}$$

$$g = \gamma^2 (1-\gamma)^2. \tag{7}$$

Kim-Kim-Suzuki (KKS) phase-field formulations were selected for the current work [36]. Use of KKS formalism provides more realistic results because it allows the composition-dependent thermodynamic and kinetic properties of coexisting phases to be evaluated at their equilibrium atom fractions rather than the bulk Zr content. The bulk Zr content and the equilibrium atom fractions in each phase are related by:

$$c = (1-h) c_{\alpha\beta} + h c_{\gamma}, \tag{8}$$

where $c_{\alpha\beta}$ and c_{γ} are the equilibrium atom fractions of Zr in their respective phases. Equation (8) illustrates why the equilibrium atom fractions of α and β must be very similar for the two-phase simplification to be valid.

KKS formalism also requires the chemical potentials of coexisting phases to be equal. This is accomplished by enforcing the following constraint:

$$\frac{\partial f_{\alpha\beta}}{\partial c_{\alpha\beta}} = \frac{\partial f_{\gamma}}{\partial c_{\gamma}}. \tag{9}$$

The gradient free energy is further defined as follows, where κ is the gradient free energy coefficient:

$$f_{grad} = \frac{\kappa}{2} |\nabla \gamma|^2. \tag{10}$$

κ and w govern the size and shape of the diffuse interface and are defined in terms of the interface width (ℓ) and interface energy (σ):

$$\kappa = 3\ell\sigma, \tag{11}$$

$$w = \frac{6\sigma}{\ell}. \tag{12}$$

ℓ and σ are used as modeling parameters in the current work. Values of $\ell = 500\,\mu$m and $\sigma = 0.5\,\text{J/m}^2$ were found to balance achievable domain size, simulation accuracy, and computational expense.

Equations 3–12 govern the relationships between the model's thermodynamic parameters. While the thermodynamic behaviors of α and β are similar, there are substantial differences in their kinetic behaviors that must be taken into account by the chemical and thermal mobilities. The chemical mobility is further defined as:

$$\begin{aligned} M_c = &(1-h_{\alpha\beta})(1-h)M_c^{\alpha} \\ &+(h_{\alpha\beta})(1-h)M_c^{\beta} \\ &+hM_c^{\gamma}, \end{aligned} \tag{13}$$

where M_c^{α}, M_c^{β}, and M_c^{γ} are the chemical mobilities in pure α, β, and γ, respectively. The thermal mobility is defined similarly in terms of the thermal mobilities in the pure phases. It is used to account for Soret effects, which have also been considered by other researchers in previous studies of constituent redistribution.

An ideal mixture approximation is used to define the chemical mobilities in pure α and β:

$$M_c^i = \frac{c_{\alpha\beta}V_m^{ref}}{RT}\tilde{D}_0^i \exp\left(\frac{-Q_i}{RT}\right), \tag{14}$$

where $\tilde{D}_0^i$ is the prefactor and Q_i is the activation energy for interdiffusion between U and Zr in phase i, and R is the ideal gas constant. Values of $\tilde{D}_0^{\alpha} = 2.0\times10^{-7}$ m^2/s, $\tilde{D}_0^{\beta} = 5.7\times10^{-5}$ m^2/s, $Q_{\alpha} = 1.7\times10^5$ J/mol, and $Q_{\beta} = 1.8\times10^5$ J/mol were selected for the current work. These values were also used by Hofman et al. and are based on an isothermal U–Zr diffusion couple experiment and self-diffusion in α-U and β-U.

Accurate representation of γ kinetics is vital because of its prevalence throughout the U–Zr system. The wide temperature and composition ranges over which γ is stable have made this a difficult task in the past. The current work accounts for the complexities of γ kinetics by incorporating temperature- and composition-dependent atomic mobilities calculated from a DICTRA assessment [37]. These are used to define the chemical mobility in pure γ according to:

$$M_c^{\gamma} = V_m^{ref}c_{\gamma}(1-c_{\gamma})[c_{\gamma}\beta_U + (1-c_{\gamma})\beta_{Zr}], \tag{15}$$

where β_U and β_{Zr} are the atomic mobilities of U and Zr in γ, respectively. These expressions are the most realistic description of γ kinetics available, and they represent a major improvement over the limited fits used in previous models.

The thermal mobilities in pure α and β are further defined as follows, where $\tilde{Q}_i^*$ is the heat of transport in phase i:

$$M_T^i = \frac{\tilde{Q}_i^* c_{\alpha\beta}\tilde{D}_0^i}{RT}\exp\left(\frac{-Q_i}{RT}\right). \tag{16}$$

Previous studies used $\tilde{Q}_{\alpha}^*$ and $\tilde{Q}_{\beta}^*$ as fitted modeling parameters. Unfortunately, no additional experimental data have been published to help define these parameters. As such, best fit values of $\tilde{Q}_{\alpha}^* = 2.0\times10^5$ J/mol and $\tilde{Q}_{\beta}^* = 4.5\times10^5$ J/mol were drawn from previous studies [15,16] for the current work and are treated as uncertain parameters.

Considerably more effort has been invested to characterize the parameters associated with the Soret effect in γ. Experimental measurements of the effective heats of transport of U and Zr in γ ($\tilde{Q}_U^*$ and $\tilde{Q}_{Zr}^*$, respectively) were made by D'Amico, Campbell, and Huntington [38,39]. Sohn et al. later contributed additional measurements collected from a ternary U-Pu-Zr alloy [13]. Estimates of the heat of transport in γ have also been made from spatial constituent gradients in PIE data [12]. The estimates were made under the assumption that the composition profiles had reached steady-state, and values typically ranged from 5.0×10^4–2.0×10^5 J/mol. Measured, estimated, and fitted values were aggregated and analyzed by Mohanty et al. [34]. The average values are used in the current work to define the thermal mobility in pure γ as follows:

$$M_T^{\gamma} = c_{\gamma}(1-c_{\gamma})(\tilde{Q}_U^*\beta_U + \tilde{Q}_{Zr}^*\beta_{Zr}). \tag{17}$$

The final two parameters required by the phase-field model are the bulk free energy functions $f_{\alpha\beta}$ and f_{γ}. While CALPHAD free energies can sometimes be incorporated directly into a phase-field model, it is often advantageous to simplify them to improve numerical performance. Care must be taken to preserve the transition temperatures, equilibrium atom fractions, phase fractions, and chemical potentials predicted by the free energies within the composition and temperature ranges of interest. In the current work, the CALPHAD free energies for α, β, and γ [26] were simplified to reduce the size of the nonlinear system and to eliminate inflection points associated with the γ_1-γ_2 miscibility gap. The resulting bulk free energy functions are given below:

$$\begin{aligned} f_{\alpha\beta} = &(1-c)G_0^{\alpha-U} + c\left(38000 + G_0^{\alpha-Zr}\right) \\ &+RT[(1-c)\ln(1-c) + c\ln c] \\ &+(1-c)c(-7000) \\ &-29.333333T + 24106.6667, \end{aligned} \tag{18}$$

$$\begin{aligned} f_{\gamma} = &(1-c)G_0^{\gamma-U} + c + G_0^{\beta-Zr} \\ &+R(1010)[(1-c)\ln(1-c) + c\ln c] \\ &+(1-c)c\{28369.0639 - 14.741714(1010) \\ &+4992.02639(1-c)c \\ &+2609.10711[(1-c)c]^2\} \end{aligned} \tag{19}$$

where $G_0^{\alpha-U}$, $G_0^{\alpha-Zr}$, $G_0^{\gamma-U}$, and $G_0^{\beta-Zr}$ are reference free energies taken from Ref. [40].

A comparison between the partial U–Zr phase diagrams predicted by the original and simplified free energies is shown in Fig. 2. The comparison shows that within the composition and temperature ranges of interest, the simplified free energies introduce only minor errors in the predicted transition temperatures, equilibrium atom fractions, and phase fractions. Simplification of the free energies increases the driving force associated with chemical potential gradients by a factor of approximately five. This effect is mitigated by dividing the chemical mobility in Eq. (1) by the same factor, restoring the original rate of evolution. Incorporation of these realistic thermodynamic behaviors represents another advantage of the current work.

2.2. Heat generation and transport components and parameters

Heat generation and transport must also be modeled to perform realistic analyses of U–Zr fuel element behavior. Heat generation is coupled to the composition because only U undergoes fission. Constituent redistribution therefore contributes to radial variations in power generation over the life of a fuel element. A composition-dependent radial power profile introduced in a previous study was used to account for these variations in the current work [16]. Heat

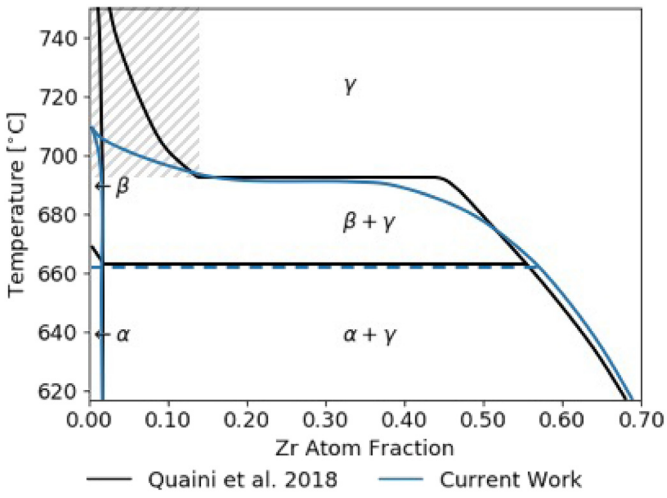


Fig. 2. Partial U–Zr phase diagrams calculated from the original [26] and simplified free energies. α-β kinetic parameters are interpolated across the transition temperature indicated by the dashed line at 662 °C. The comparison shows that the phase fields predicted by the free energies are in good agreement except for within the region marked with gray hashes. Discrepancies in this region do not bias the model's predictions because this region does not have a significant role in constituent redistribution. The simplification introduces only minor errors into the predicted transition temperatures, equilibrium atom fractions, and phase fractions outside of this region.

generation decreases linearly from its initial value at a rate of 1% per at% burnup to account for changes in U content.

Solid and gaseous fission products are produced and deposited into the fuel matrix over time, causing the fuel to swell. The majority of swelling at the beginning of life is due to gaseous fission products, which migrate to form bubbles. The bubbles grow, interconnect, and ultimately provide paths for fission gas release into the fuel-cladding gap and plenum [41]. The low thermal conductivity of the fission gases degrades the fuel's overall thermal conductivity, increasing the temperature gradient across the fuel [42–44]. Swelling results in contact between the fuel and cladding, and anisotropic swelling in low-temperature phases produces grain boundary tearing and fuel damage.

Porosity appears to develop more uniformly at the beginning of life but correlate more strongly with phase composition as burnup increases [41]. Estimates of radial porosity at intermediate and high burnups suggest that porosity development continues in some regions and stalls or possibly even reverses in others [45]. Mechanistic swelling models based on those used for oxide fuels are still under development.

The gap between the fuel and cladding is filled with liquid sodium during manufacturing to enhance heat transfer. The fission gas channels and grain boundary tears provide paths for the sodium to infiltrate the fuel. The high thermal conductivity sodium enhances the overall thermal conductivity of the fuel, mitigating the effects of the remaining fission gas [45]. The total amount of liquid sodium deposited in the fuel has been calculated from changes in fuel element dimensions and plenum sodium level. However, the spatial distribution of the sodium in the fuel is not well known.

While nuclear instruments and heat output can be used to calculate reactor power, much of what we know about the fuel's temperature and thermal conductivity under irradiation has been inferred from the locations of phase boundaries observed in PIE data [45]. Optimization of phase transition temperatures in some of the previous studies suggests that there is a problem with this approach. Either the phase transition temperatures in the irradiated fuel are not known with accuracy, or these models' physical descriptions of the system were lacking. In spite of these observations, the community's knowledge of porosity development and sodium infiltration are insufficient to improve on existing thermal conductivity models at this time.

The complexity of the phase-field model components and the considerable uncertainties associated with the processes discussed above justify implementation of a simplified model for heat transport. The current work accounts for the effects of swelling, fission gas release, and sodium infiltration similarly to Hofman et al. by applying a multiplicative thermal conductivity correction factor (k_c) [42,43] to the thermal conductivity of the fresh fuel [46].

Hofman et al. noted that k_c begins to decrease from one at the beginning of life due to swelling. k_c reaches a minimum of about 0.50 (k_{swell}) around 0.75 at% burnup (BU_{swell}), when fission gas release arrests and begins to reverse the thermal conductivity degradation. Infiltration of sodium then causes k_c to increase to a steady-state value of about 0.70 (k_{infil}) around 1.50 at% burnup (BU_{infil}). k_c is applied uniformly throughout the fuel in the current work, and the transitions between maximum, minimum, and steady-state values are assumed to be linear.

While this is a substantial simplification, it is qualitatively consistent with observed fuel element behaviors. Some studies have varied k_c spatially, but we do not believe that the current understanding of porosity development and sodium infiltration are complete enough to successfully implement such an approach. In this work, we acknowledge that the swelling parameters (k_{swell} and BU_{swell}) and sodium infiltration parameters (k_{infil} and BU_{infil}) are uncertain and study their influence using sensitivity analysis to gain insight into the interactions between these behaviors and constituent redistribution.

DP-81 and DP-11 geometry, power, and surface temperature data were taken from Hofman et al. Each fuel rod had an initial radius of 2170 μm. PIE of DP-81 was conducted in a region with a linear power (q') of 240 W/m, a fuel surface temperature (T_s) of 627 °C, and a local burnup of 3.6 at%. PIE of DP-11 was conducted in a region with $q' = 250$ W/m, $T_s = 637$ °C, and a local burnup of 7.7 at %. Along with those introduced in Subsec. 2.1, these values and relationships provide all of the information necessary to quantitatively predict the dynamic coevolution of phases, constituents, and temperature throughout the two U–Zr fuel elements.

3. Methods

Simulations were conducted using MARMOT, a mesoscale fuel performance code based on the MOOSE Framework [47–49]. MARMOT uses implicit time stepping with finite element discretization, which allow for simulation of lengthy processes and complex geometries. This gives the model developed in the current work a distinct advantage over those developed in previous studies, many of which utilized explicit finite difference. The available PIE data are reported as constituent composition versus fuel slug radius at a single axial and azimuthal position. As such, 1D simulations are used in the current work to obtain the best comparison with these data. However, the model is fully capable of conducting 2D and 3D simulations without additional modifications. This will allow the model to be used for future 2D and 3D studies of different fuel geometries and higher resolution PIE data as they are made available.

A 2170 μm long, 1D, axisymmetric domain discretized into 1000 elements was used to represent DP-81 and DP-11. The left and right sides of the domain were chosen to represent the centerline and surface of the fuel, respectively. Zero-flux boundary conditions were applied to all variables at the centerline to create a symmetry plane. Zero-flux boundary conditions were also applied to all composition variables at the fuel surface, and the fuel surface

temperature was fixed using the values given in Subsec. 2.2. All simulations were conducted using an adaptive time stepping scheme with time steps ranging from 1 to 1×10^5 s.

The model includes the most realistic thermodynamic and kinetic descriptions available for U–Zr but still employs approximations, and there are still considerable uncertainties associated with many of the model's parameters. As such, the need for optimization was anticipated from the start. Optimization was performed by coupling MARMOT to DAKOTA [50] and is discussed in Sec. 4 along with model validation. Optimized parameters were used to interpret proposed behaviors such as irradiation-enhanced diffusion. Sensitivity analysis was then conducted to identify the most influential parameters and deconvolute complex metallic fuel behaviors by examining the interactions between swelling, sodium infiltration, and constituent redistribution. These efforts are covered in Sec. 5.

4. Model optimization and validation

The experimental Zr data were captured graphically from the plots published by Hofman et al. The experimental composition data were originally collected using an electron microprobe to sample 40×40 μm regions of the fuel cross-section, yielding the average Zr composition at various points [12]. The data exhibit considerable scatter, even in the single-phase γ region. Visual inspection of the cross-section [25] shows that the fuel exhibited minor but significant azimuthal variations in its structure. It is unclear whether the microprobe data are statistically representative of these variations. This could potentially lead to conflicts with the model, which predicts the average azimuthal behavior, but the available PIE data are insufficient to resolve this issue at this time. We recommend gathering composition data from entire fuel cross-sections in the future. These data will allow refinement of constituent redistribution models and improve estimates of porosity and fuel damage.

A simulation was first conducted for DP-81 using the parameter values given in Sec. 2. This simulation provided a baseline for optimization and was used to qualitatively evaluate the model's physics. The predicted Zr, phase, and temperature profiles are shown in Fig. 3 alongside the experimental Zr profile. The results show that the temperature profile has a reasonable shape and that the predicted temperatures fall within the expected range. The fuel transitions from γ to β+γ to α+γ as temperature decreases from the centerline, and the phase transitions occur at about 693°C and 662°C, as expected. Zr content has increased slightly in the γ region, decreased slightly in the β+γ region, and remained fairly constant at its initial value of 22 at% in the α+γ region. These features are consistent with the general understanding of constituent redistribution, but the model clearly does not predict evolution on the scale observed in the experiments. These observations suggest that the model's physics are capturing the expected behaviors, but optimization is necessary to account for parameter uncertainties, simplifications, and/or the effects of irradiation.

Twelve uncertain parameters were identified in the model, including six kinetic parameters ($\tilde{D}_0^{\alpha}$, $\tilde{D}_0^{\beta}$, M_c^{γ}, $\tilde{Q}_{\alpha}^{*}$, $\tilde{Q}_{\beta}^{*}$, M_T^{γ}), four parameters associated with k_c (k_{swell}, BU_{swell}, k_{infil}, and BU_{infil}), and two operational parameters (q' and T_s). Each parameter was multiplied by a scalar modifier with a nominal value of one to establish an interface for optimization and sensitivity analysis. The M_c^{γ} and M_T^{γ} modifiers are associated with uncertainties in the atomic mobilities and effective heats of transport of U and Zr in γ, respectively.

Ranges were established for each of the modifiers based on recommended uncertainties from the literature, lessons drawn from previous studies, and in the absence of either of these, physical intuition. Parameters associated with the constituent mobility were permitted to increase by a factor of up to ten to explore the range of optimized values identified by previous studies. The parameters associated with the thermal mobility, which have traditionally been used as modeling parameters, were allowed to vary by up to 50%. This value was chosen somewhat arbitrarily because of the parameters' large, poorly characterized uncertainties. k_c parameters and q' were permitted to vary by up to 5% based on discussions in Refs. [42,45], respectively. Finally, T_s was allowed to vary by up to 1% (about ±9°C). This variation is on the order of the ±5°C recommended in Ref. [45]. The extra variation allows exploration of the temperatures above the region where the δ phase is stable. The uncertain parameters and their modifier ranges are summarized in Table 1.

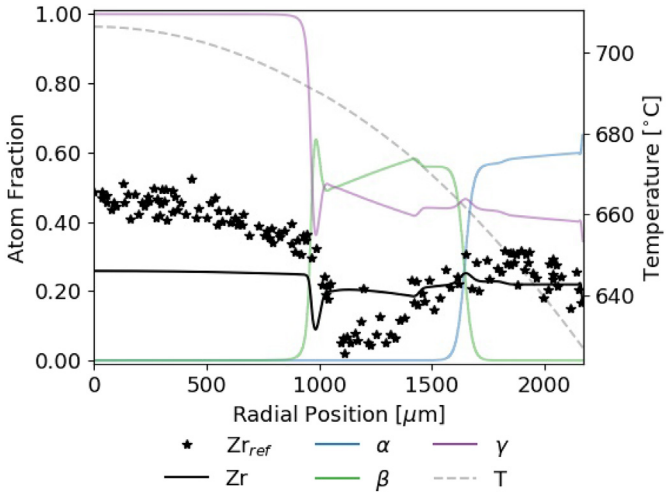


Fig. 3. The Zr, phase, and temperature profiles predicted for DP-81 prior to optimization. The temperature profile has a reasonable shape, and the temperatures are within the expected range. The phase fields and phase transition temperatures are consistent with experimental observations and the U–Zr phase diagram. Zr content has increased slightly in the γ region, decreased slightly in the β+γ region, and remained fairly constant in the α+γ region. The model under predicts constituent redistribution, but its physics appear to capture the expected behaviors. Uncertainties in model parameters, simplifications, and/or the effects of irradiation make optimization necessary. Graphically captured experimental composition data, which were originally collected using an electron microprobe, are included for comparison.

A smoothed fit for the DP-81 data was developed to facilitate optimization, and the DP-11 dataset was reserved as an independent validation case. DAKOTA was used to identify the modifier values that yielded the best fit between the predicted and experimental Zr profiles. While the ranges established for all of the modifiers were physically reasonable, inclusion of too many modifiers in the optimization would have increased the risk of overfitting. This risk was mitigated by optimizing as few modifiers as possible. Using previous studies as guides, modifiers were introduced in the following order: γ kinetic parameters, β kinetic parameters, α kinetic parameters, k_c parameters, and the operational parameters, q' and T_s. Improvements diminished after introducing only four parameters, concluding the optimization. The chosen modifiers and their optimized values are given in Tab. 2. The remainder of this section focuses on the results obtained using the optimized model. The effects of uncertainties in the twelve uncertain parameters are examined further using sensitivity analysis in Sec. 5.

As in previous studies, diffusivities needed to be enhanced to yield predictions consistent with experimental observations. This supports the hypothesis that diffusion is being enhanced by irradiation. Enhancement mechanisms such as vacancy production and

Table 1
The names, values, references, and modifier ranges for the twelve uncertain parameters considered during optimization and sensitivity analysis. M_c^γ and M_T^γ values are not given because they vary with the composition- and temperature-dependent atomic mobilities of U and Zr in γ. Modifier ranges were selected based on recommended uncertainties from the literature and lessons drawn from previous studies. Note that the DP-11 dataset was excluded from model optimization, allowing it to be used as an independent validation case.

Parameter	Value	Ref.	Minimum Modifier	Maximum Modifier
$\tilde{D}_0^\alpha$	2.0×10^{-7} m^2/s	[12]	1.00	10.0
$\tilde{D}_0^\beta$	5.7×10^{-5} m^2/s	[12]	1.00	10.0
M_c^γ	$M_c^\gamma(c,T)$ m^5/(J s)	[37]	1.00	10.0
$\tilde{Q}_\alpha^*$	2.0×10^5 J/mol	[15,16]	0.50	1.50
$\tilde{Q}_\beta^*$	4.5×10^5 J/mol	[15,16]	0.50	1.50
M_T^γ	$M_T^\gamma(c,T)$ m^2/s	[34]	0.50	1.50
k_{swell}	0.50	[12]	0.95	1.05
BU_{swell}	0.75 at%	[12]	0.95	1.05
k_{infil}	0.70	[12]	0.95	1.05
BU_{infil}	1.50 at%	[12]	0.95	1.05
q'	240 W/m (DP-81)	[12]	0.95	1.05
T_s	627°C (DP-81)	[12]	0.99	1.01

Table 2
Optimized modifier values, which yield the best fit between the predicted and experimental Zr profiles for DP-81. Optimization was conducted with as few unitless modifiers as possible to avoid over-fitting. Good agreement required that diffusivities be enhanced, but no modification of the phase transition temperatures was necessary.

Modifier	Optimized Value
$\tilde{D}_0^\beta$	2.722
M_c^γ	9.981
$\tilde{Q}_\beta^*$	0.617
M_T^γ	1.395

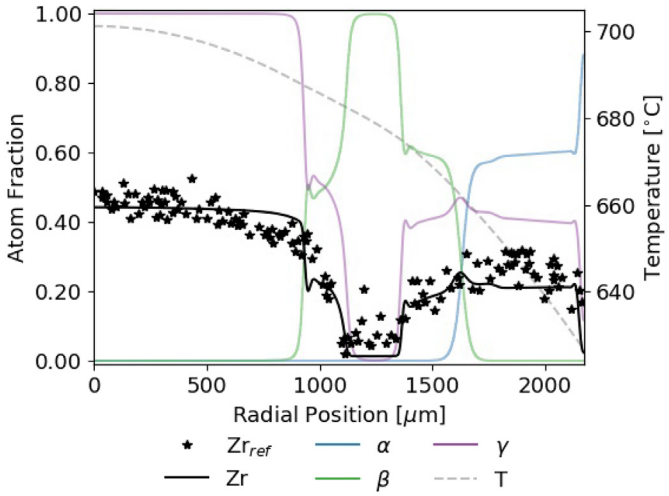


Fig. 4. The composition and temperature profiles predicted for DP-81 after optimization. While the temperature profile has changed little, the phase and Zr profiles differ substantially from those predicted prior to optimization. The model predicts the formation of an intermediate region rich in β and depleted of Zr (referred to as a Zr bathtub). Zirconium displaced from the bathtub has been driven toward the fuel centerline and peaks near the expected value. The predicted Zr profile is in good agreement with the experimental data.

Pu generation have been proposed, but these contributions are convoluted with uncertainties in the unirradiated kinetic parameters. Modification of the phase transition temperatures was not necessary in the current work, suggesting that other models' thermodynamic descriptions were inadequate or that the phase diagrams from which they were derived are uncertain.

The composition and temperature profiles predicted for DP-81 using the optimized modifiers are shown in Fig. 4. The temperature profile is again consistent with expectations. The shape of the profile has changed slightly due to the effects of more substantial constituent redistribution on the thermal conductivity. The phase profiles differ from those shown in Fig. 3 due to formation of a β-rich intermediate region. This region has been depleted of nearly of its Zr, forming what has been traditionally referred to as a Zr bathtub. Zirconium from the bathtub appears to have been driven toward the fuel centerline, which is consistent with previous studies and experimental observations. Zirconium peaks near the expected value at the fuel centerline, and there is good overall agreement between the predicted and experimental Zr profiles.

The model was then validated against PIE data from DP-11, and the predicted composition and temperature profiles are shown in Fig. 5. The predicted phase boundaries correspond very well to the edges of the experimental Zr bathtub. The model predicts inward migration of Zr, and the agreement between the predicted and experimental Zr profiles is good overall. However, several discrepancies between the Zr profiles warrant further discussion and investigation. The model slightly over-predicts the Zr atom fraction in γ, and agreement is poorer within the Zr bathtub, particularly near its inner and outer edges. Recall that DP-11 reached more than twice the burnup of DP-81 and that uncertainties associated with swelling and sodium infiltration increase with burnup. These uncertainties likely contribute to inconsistencies between the shapes, widths, and positions of the predicted and experimental Zr bathtubs. Accurate prediction of constituent redistribution using the same parameters across multiple fuel elements has been a challenge in the past [16]. This level of agreement is therefore very promising. It suggests that the model adequately represents the necessary physics and material properties. Interestingly, the composition and temperature profiles continued to evolve until the final time step for both fuel elements. This suggests that neither had reached steady-state, and that we may have more to learn from elements taken to even higher burnups. Further optimization and validation efforts will be conducted as additional PIE data is made available.

With some confidence in the model's predictions, it can now be used to learn about the details of the material's behavior. DP-81 profiles predicted at various burnups are shown in Fig. 6. These provide valuable insights into how constituent redistribution occurs over time. The results show that the majority of Zr evolution

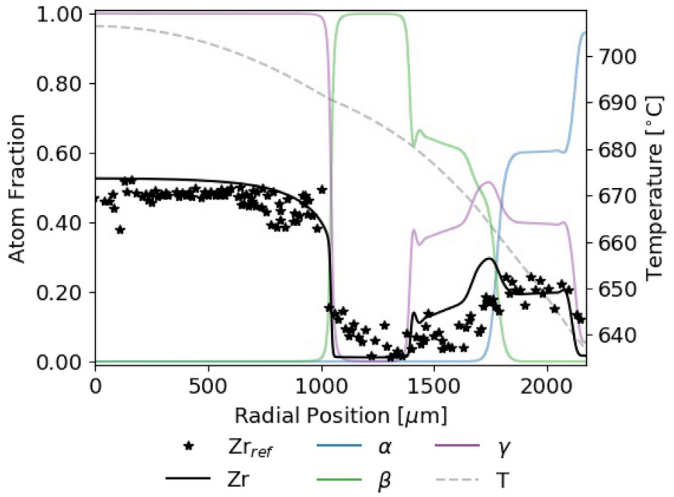
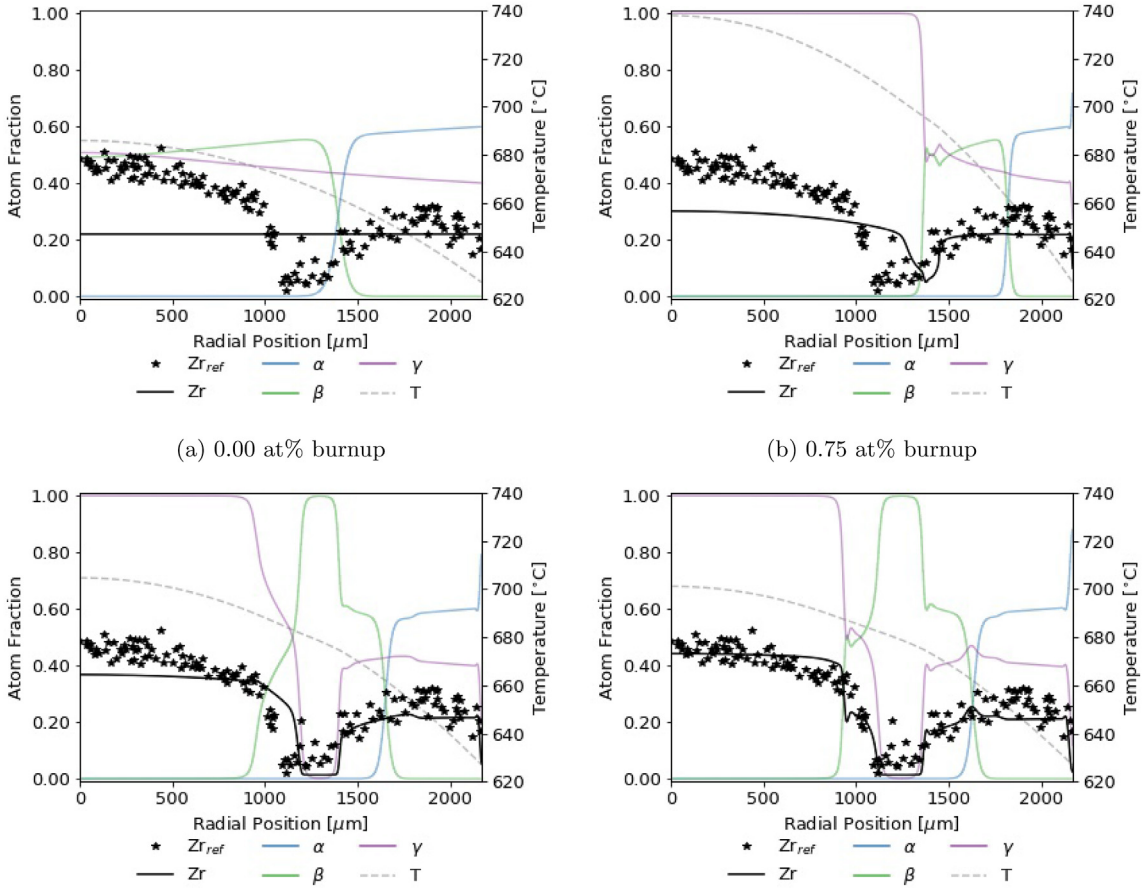


Fig. 5. The composition and temperature profiles predicted for DP-11 after optimization. Despite additional uncertainties associated with the fuel element's higher burnup, the overall agreement between the predicted and experimental Zr profiles is favorable. This suggests that the model adequately captures the necessary physics and material properties.

occurs within the γ and β+γ regions, where temperatures and diffusivities are highest. The two regions possess heats of transport with opposite signs, and the profiles show that the interface between the two regions moves over time due to variations in the thermal conductivity. The behaviors shown in Fig. 6b suggest that k_{swell} and BU_{swell} influence the location of the outer edge of the Zr bathtub. The location of the inner edge of the Zr bathtub appears to be similarly affected by k_{infil} and BU_{infil}, as shown in Fig. 6c. Centerline temperature is predicted to peak at about 740°C immediately before fission gas release, which is in fair agreement with the 720°C predicted by Hofman et al. Diffusion seems to scale the magnitude of constituent redistribution. It is enhanced by the high temperatures induced by swelling and reduced by sodium infiltration. These behaviors effectively illustrate the complexity of the coupled, kinetic processes at work in metallic fuels under irradiation.

The model's ability to capture these behaviors is largely due to its description of γ kinetics, which incorporates all of the complexities of the DICTRA atomic mobilities. More familiar parameters can be calculated from the model for comparison to those used in previous studies. The interdiffusion coefficient between U and Zr in γ ($\tilde{D}_\gamma$) and the heat of transport in γ ($\tilde{Q}^*_\gamma$) are calculated as follows, where Eq. (20) uses an ideal mixture approximation similar to that used to derive Eq. (14):



(a) 0.00 at% burnup

(b) 0.75 at% burnup

(c) 1.50 at% burnup

(d) 3.60 at% burnup

Fig. 6. Composition and temperature profiles predicted for DP-81 at various burnups. Phase fields in the fuel shift from their initial state (a) as temperature increases due to swelling. Centerline temperature peaks immediately before fission gas release (b). Temperatures then decrease and a recognizable Zr bathtub forms as sodium infiltrates the fuel (c). These behaviors suggest that the final constituent redistribution profiles (d) are strongly influenced by diffusion in the γ and β+γ regions. Movement of the interface between the two regions in response to changes in thermal conductivity affects the location of the Zr bathtub, and the rate of diffusion governs the magnitude of constituent redistribution.

$$\tilde{D}_\gamma = \frac{RT}{V_m^{ref} c_\gamma} M_c^\gamma, \tag{20}$$

$$\tilde{Q}_\gamma^* = \frac{M_T^\gamma}{M_c^\gamma} V_m^{ref}. \tag{21}$$

$\tilde{D}_\gamma$ and $\tilde{Q}_\gamma^*$ values calculated from the model's DP-81 predictions are shown in Fig. 7. Fig. 7a and b shows how the temperature- and composition-dependent parameters vary across the fuel's radius at the final time step. Radial $\tilde{D}_\gamma$ and $\tilde{Q}_\gamma^*$ values were averaged at each time step for comparison to previous studies. The evolution of the average parameters with burnup is illustrated in Fig. 7c and d, where dotted lines and labels have been added to identify BU_{swell}, BU_{infil}, and the dominant behavior influencing k_c at each stage of burnup. The magnitude of $\tilde{D}_\gamma$ is reasonable throughout the fuel, and $\tilde{Q}_\gamma^*$ is in good agreement with the values of −50 to −100 kJ/mol recommended by Hofman et al. The average $\tilde{D}_\gamma$ is predicted to vary by almost an order of magnitude, and the average $\tilde{Q}_\gamma^*$ is predicted to vary by nearly 30% over the life of the fuel element.

The simplified thermodynamic and kinetic descriptions employed in previous studies may have inhibited their ability to adequately resolve these behaviors [11–16]. Optimization of additional parameters in the previous studies, such as the parameters associated with α kinetics, may have been needed to compensate for the simplified descriptions or may not have been needed at all. The current work demonstrates that comparable levels of accuracy can be obtained by optimizing only four parameters without ignoring the effects of coupled irradiation behaviors. Nonetheless, uncertainties in the available material property data and the relationships between different fuel behaviors remain. Without further study, these uncertainties will continue to limit achievable model accuracy. Sensitivity analysis is applied in the next section to investigate these uncertainties, refine the physical understanding of the system, and provide guidance for future experiments and modeling efforts.

5. Sensitivity analysis

Sensitivity analysis was applied to DP-81 to identify the most influential model parameters and to gain further insight into the physics underlying constituent redistribution. This information will guide researchers toward the most impactful areas, increasing the overall effectiveness of future experiments and modeling efforts by enabling efficient allocation of research time and funding.

Sensitivity analysis was conducted by sampling random modifier values from uniform distributions based on the ranges given in Table 1. The first set of simulations involved varying each of the

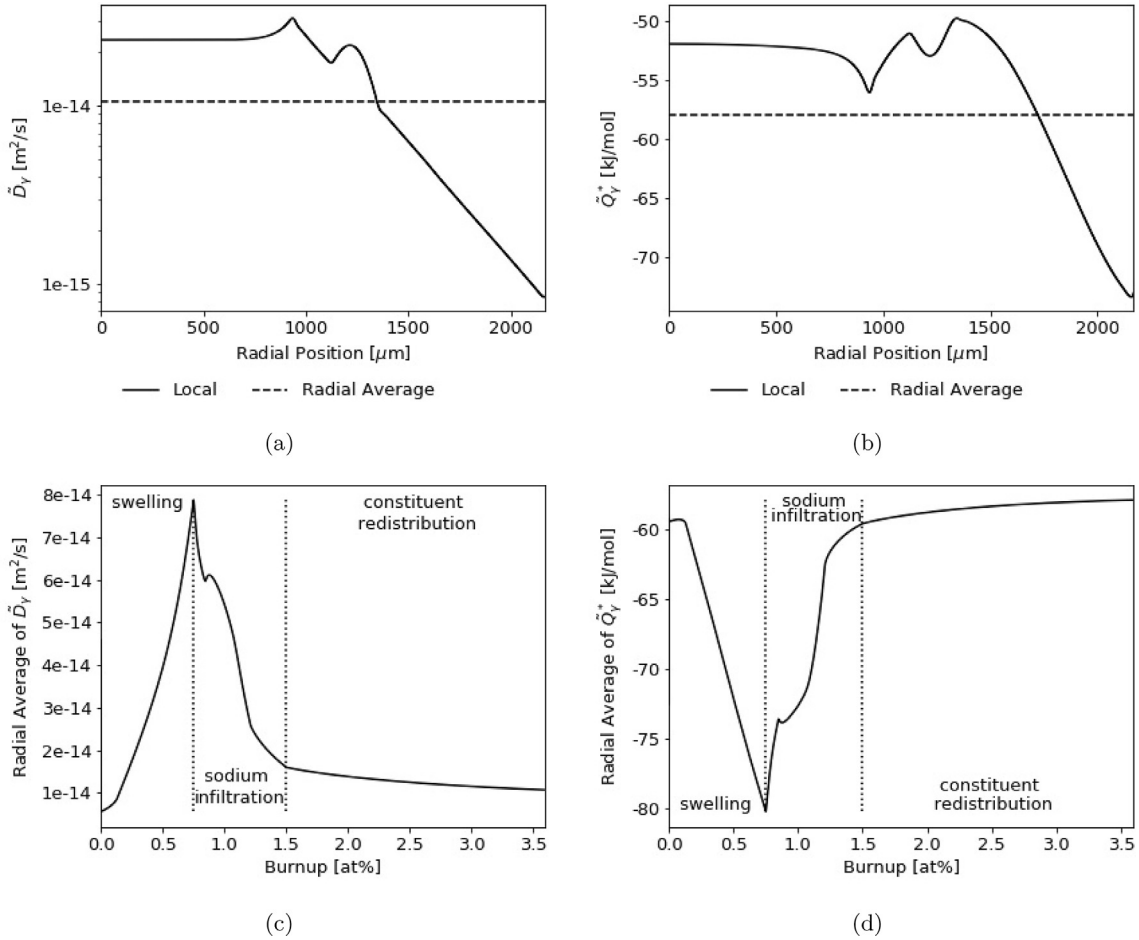


Fig. 7. $\tilde{D}_\gamma$ and $\tilde{Q}_\gamma^*$ values calculated from the model's DP-81 predictions. Variations in the parameters across the fuel radius at the final time step and their radial averages are shown in (a) and (b). The average values are shown to vary with burnup in (c) and (d), where dotted lines and labels identify BU_{swell}, BU_{infil}, and the dominant behavior influencing k_c at each stage of burnup. The magnitude of $\tilde{D}_\gamma$ is reasonable, and its average varies by almost an order of magnitude over the life of the fuel element. The average $\tilde{Q}_\gamma$, which is predicted to vary by about 30% over the fuel element's life, is in good agreement with the values of −50 to −100 kJ/mol recommended by Hofman et al. These results help to illustrate the complexities captured by the model's atomic mobility-based description of γ kinetics.

model's twelve uncertain parameters one at a time. Twenty simulations were conducted using different modifier values for each parameter. The remaining modifiers were held constant at their optimized values (or their original values, if the modifier was not included in the optimization).

Visual comparisons between the resulting Zr profiles are made in Subsecs. 5.1–5.3. These provide qualitative insights into the effects of the coupled fuel behaviors and the interactions between them. Variations were quantified by calculating four scalar response functions, including the centerline temperature (T_{CL}), centerline Zr atom fraction (Zr_{CL}), location between the γ and $\beta+\gamma$ regions ($x_{\gamma\beta}$), and location between the $\beta+\gamma$ and $\alpha+\gamma$ regions ($x_{\beta\alpha}$).

Linear sensitivities were fitted to each response function, and their slopes were normalized to account for differences in modifier range size and response function magnitude. The resulting relative sensitivities, which illustrate how the predicted composition and temperature profiles vary with each parameter, are shown in Fig. 8. The results indicate that α kinetic parameters have little effect on the model's predictions while β and γ kinetic parameters are more impactful. k_{swell} and k_{infil} appear to be the most influential k_c parameters, and the operational parameters are shown to have substantial effects. T_s was the most influential parameter overall (even more than the γ kinetic parameters). These observations are explored in greater detail in the following subsections.

Varying one parameter at a time yields intuitive, easily interpreted results, but this method cannot be used to detect interactions between the parameters. A second set of 200 simulations was therefore conducted to search for possible correlations. The simulations involved varying $\tilde{D}_0^\beta$, M_c^γ, $\tilde{Q}_\beta^*$, M_T^γ, k_{swell}, k_{infil}, q', and T_s simultaneously. $\tilde{D}_0^\alpha$, $\tilde{Q}_\alpha^*$, BU_{swell}, and BU_{infil} were held constant at their original values due to their limited influence on the model's predictions. Pearson and Spearman correlation coefficients were calculated from the results and showed no significant linear or nonlinear correlations between the parameters.

5.1. Sensitivity to kinetic parameters

As anticipated, model predictions were found to be insensitive to variations in $\tilde{D}_0^\alpha$ and $\tilde{Q}_\alpha^*$. α is stable only near the periphery of the fuel, where temperatures fall below 662°C. Despite its abundance in this region, its relatively slow kinetics restrict constituent migration. Even a ten-fold increase in α diffusivity is insufficient to appreciably affect constituent redistribution. These findings support those of Hofman et al. who predicted that fuels operated below 662°C would not experience significant constituent redistribution. This explains why optimization of the α kinetic parameters was unnecessary in the current work.

The Zr profiles predicted using twenty different $\tilde{D}_0^\beta$ values are shown in Fig. 9. The profile obtained using the optimized $\tilde{D}_0^\beta$ value is shown in black. Red and blue identify the profiles obtained using the maximum and minimum values of $\tilde{D}_0^\beta$, respectively. These are sometimes obscured by the black profile if the optimized value lies near one of its range's extremes. The remaining profiles are shown in gray, indicating that they were obtained using intermediate values of $\tilde{D}_0^\beta$. These often form solid gray regions due to the large number of profiles.

The results show that the profiles differ at the boundaries of the $\beta+\gamma$ region. Raising $\tilde{D}_0^\beta$ promotes Zr transport from the inner edge of the Zr bathtub to the peak between the $\beta+\gamma$ and $\alpha+\gamma$ regions. This peak also seems to appear in the experimental results, but large $\tilde{D}_0^\beta$ values unrealistically inflate its size. Recall that the thermal conductivity of Zr is lower than that of U and that regions richer in U generate more heat. Increasing $\tilde{D}_0^\beta$ therefore increases heat generation in the fuel's intermediate region and decreased thermal

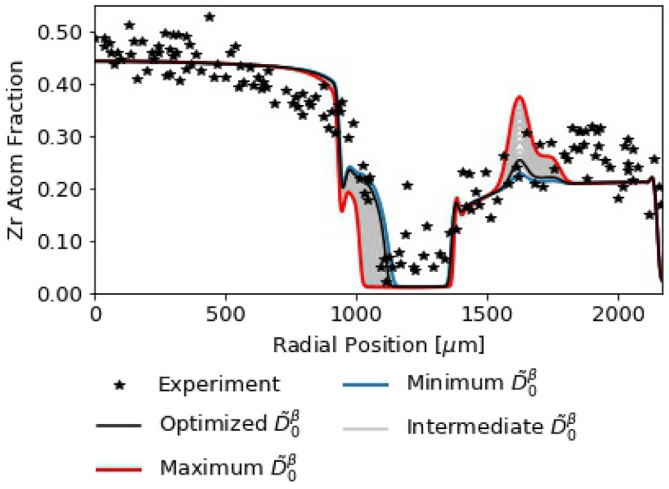


Fig. 9. Zr profiles predicted for DP-81 using twenty different $\tilde{D}_0^\beta$ values. Black, red, and blue identify the profiles obtained using the optimized, maximum, and minimum values of $\tilde{D}_0^\beta$, respectively. Profiles obtained using intermediate $\tilde{D}_0^\beta$ values are shown in gray. Increasing $\tilde{D}_0^\beta$ promotes Zr transport from the inner edge of the Zr bathtub to the peak between the $\beta+\gamma$ and $\alpha+\gamma$ regions. Large $\tilde{D}_0^\beta$ values unrealistically inflate the size of the peak. Model accuracy is believed to be limited by the lack of composition dependence in $\tilde{D}_0^\beta$.

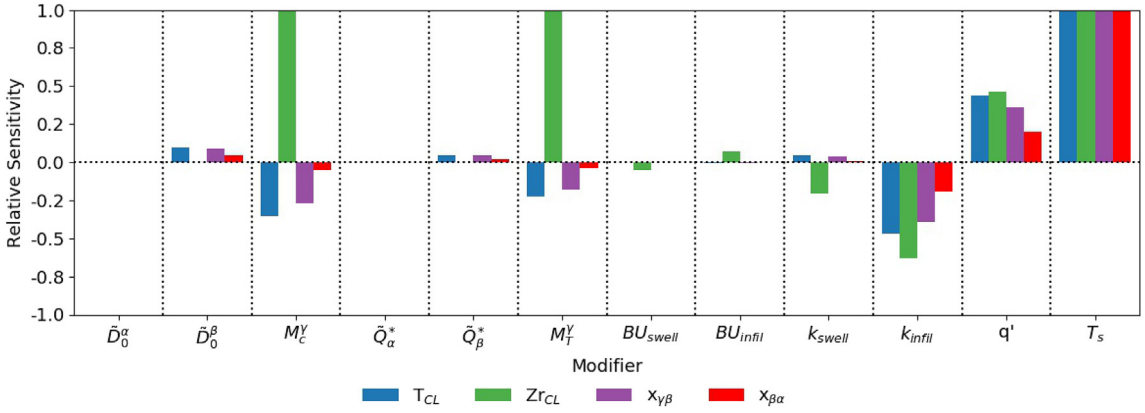


Fig. 8. The relative sensitivity of each scalar response function to variations in the model's twelve uncertain parameters. The results have been normalized to account for differences in modifier range size and response function magnitude. The predicted composition and temperature profiles are most sensitive to the β and γ kinetic parameters, k_{swell}, k_{infil}, q', and T_s.

conductivity near its surface. This results in a modest rise in T_{CL}, as shown in Fig. 8. The $\tilde{D}_0^\beta$ value used by Hofman et al. and in the current work is based on a single isothermal diffusion couple experiment. The parameter lacks any composition dependence and likely limits the achievable accuracy of the current work. Additional diffusion couple experiments will be necessary to improve the understanding of β kinetics.

The Zr profiles predicted using different M_c^γ values are shown in Fig. 10, where they have been color coded as before for consistency. The shape of the blue profile resembles the pre-optimization results shown in Fig. 3, supporting the argument that diffusion scales the magnitude of constituent redistribution. Increasing the M_c^γ modifier enhances Zr migration from the intermediate region toward the fuel centerline, promoting formation of the Zr bathtub. Fig. 8 shows that inward Zr migration lowers T_{CL}, as expected. Note that the radial geometry of the fuel element is responsible for the apparent mismatch between Zr depletion in the bathtub and Zr enrichment in the central region.

Sensitivity analysis results for $\tilde{Q}_\beta^*$ are shown in Fig. 11. The Zr profiles show that increasing $\tilde{Q}_\beta^*$ promotes outward Zr migration. Variations in $\tilde{Q}_\beta^*$ and $\tilde{D}_0^\beta$ have similar effects on the model's predictions.

Finally, sensitivity analysis results for M_T^γ are shown in Fig. 12. As was found with the β kinetic parameters, varying M_T^γ and M_c^γ have similar effects.

The temperature profile, which dictates the location of phase boundaries within the fuel, is affected by constituent redistribution, heat generation, and heat transport. The results presented in Fig. 6 of Sec. 4 show that the latter two processes dominate. The phases' kinetic parameters govern the migration of constituents within and between the various phase fields, scaling the magnitude of constituent redistribution. This contributes to minor shifts in the phase boundaries due to differences in the phases' solubilities and the thermal conductivity's composition dependence. The model's predictions are most sensitive to the β and γ kinetic parameters. These dictate the depth and shape of the Zr bathtub by contributing to outward and inward Zr migration, respectively. Both sets of parameters must be known with accuracy to predict constituent redistribution in U–Zr fuels, but only γ kinetic parameters have been studied in any detail. For these reasons, we recommend targeting β kinetic parameters in future experiments and modeling efforts.

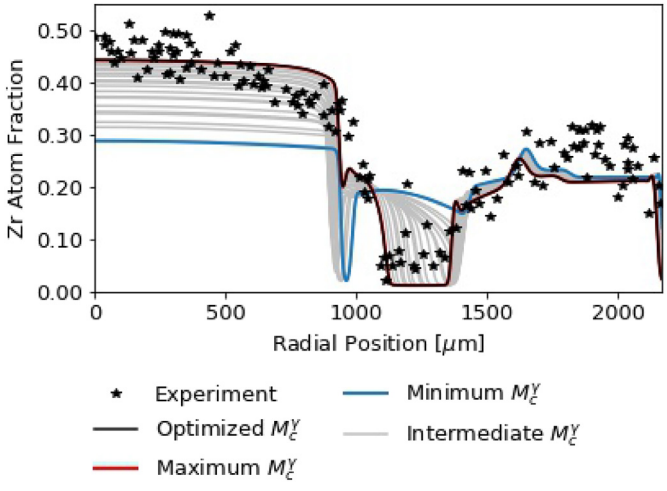


Fig. 10. Zr profiles predicted using different M_c^γ values. Increasing the M_c^γ promotes inward migration of Zr from the bathtub region, supporting the observation that diffusion scales the magnitude of constituent redistribution.

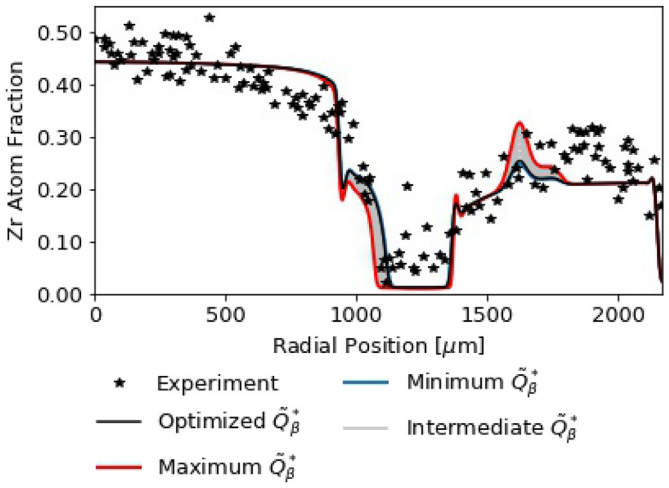


Fig. 11. Zr profiles predicted using different $\tilde{Q}_\beta^*$ values. The model's predictions respond similarly to variations in the $\tilde{Q}_\beta^*$ and $\tilde{D}_0^\beta$.

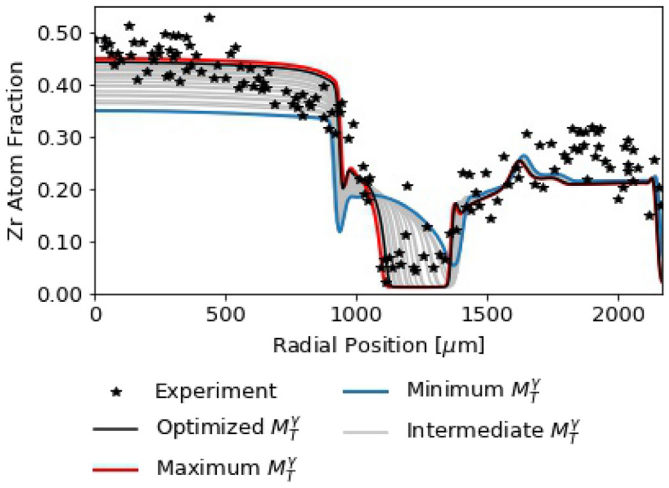


Fig. 12. Zr profiles predicted using different M_T^γ values. Variations in M_T^γ and M_c^γ have similar effects on the model's predictions.

5.2. Sensitivity to the thermal conductivity correction parameters

Sensitivity analysis was then applied to the parameters used to calculate k_c. Variations in BU_{swell} and BU_{infil} produced modest variations in $x_{\gamma\beta}$, but the resulting Zr profiles were almost indistinguishable. This suggests that a sufficient amount of evolution had occurred after these transitions to mask the variations' effects. These parameters may be more important earlier in the fuel element's life, but constituent redistribution at low burnups is not significant enough to warrant further investigation at this time. Variations in k_{swell} and k_{infil} had significant effects on the model's predictions, which help to clarify the influence of swelling, fission gas release, and sodium infiltration on constituent redistribution.

Recall that k_c has a value of one at the beginning of life and decreases to simulate the formation of porosity within the fuel. It reaches a minimum defined by k_{swell} at a burnup of BU_{swell} before increasing due fission gas release and sodium infiltration. Changing temperatures cause the interface between the γ and $\beta+\gamma$ regions to traverse the fuel radially. Fuel temperatures peak at BU_{swell}, which enhances constituent evolution, particularly in the vicinity of the interface. The Zr profiles predicted using different k_{swell} values are

shown in Fig. 13. The results suggest that variations in k_{swell} affect the location of the Zr bathtub. Smaller values of k_{swell}, which correspond to higher peak fuel temperatures, shift the Zr bathtub outward.

The steady-state value of k_c is given by k_{infil}. k_{infil} determines the fuel's long-term temperature profile and therefore the final location of the interface between the γ and $\beta+\gamma$ regions. The Zr profiles predicted using different k_{infil} values are shown in Fig. 14. The results show that variations in k_{infil} primarily alter the shape and location of the left edge of the Zr bathtub. Larger k_{infil} values, which correspond to higher thermal conductivities and lower temperatures, slow evolution and produce rougher features. Smaller k_{infil} modifiers raise temperatures, enabling diffusion to smooth rough features in the Zr profiles. These observations reinforce the idea that constituent redistribution is a kinetic process. Prediction of a fuel element's final Zr profile therefore requires that its entire life be taken into account.

Sensitivity analysis shows that the k_c parameters' influence on heat transport through the fuel affects the shape and location of the Zr bathtub. The uniform k_c applied in the current work yields more favorable agreement for DP-81, suggesting that this approximation is satisfactory for burnups lower than 3.6 at% and perhaps beyond. However, we noted in Subsec. 2.2 that porosity appears to correlate more strongly with the phase boundaries as burnup increases. This behavior is likely related to variations in the phases' crystal structures, mechanical properties, and transport properties. Development of a spatially varying thermal conductivity correction factor may be necessary to improve model accuracy for higher burnup fuel elements like DP-11. To this end, we recommend further study of porosity development and evolution both before and after fission gas release. We also recommend that resources be devoted to characterizing the spatial distribution of logged sodium in the fuel so that more accurate temperature distributions can be calculated.

5.3. Sensitivity to operational parameters

Finally, sensitivity analysis was applied to study the effects of variations in q' and T_s on the model's predictions. Uncertainties associated with neutronic and coolant calculations make it difficult to define these parameters with absolute certainty, but even small

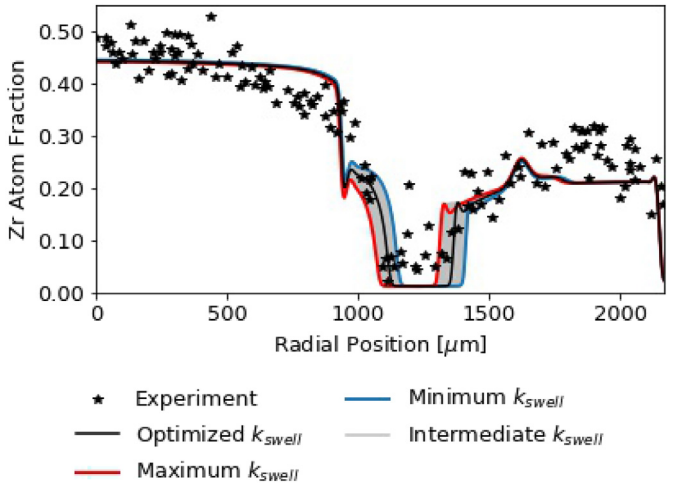


Fig. 13. Zr profiles predicted using different k_{swell} values. k_{swell} dictates the minimum thermal conductivity of the fuel and is related to the fuel's total porosity. Variations in k_{swell} influence fuel temperatures, kinetics, and the location of the interface between the γ and $\beta+\gamma$ regions, which traverses the fuel radially as the temperatures change. The results show that variations in k_{swell} affect both edges of the Zr bathtub.

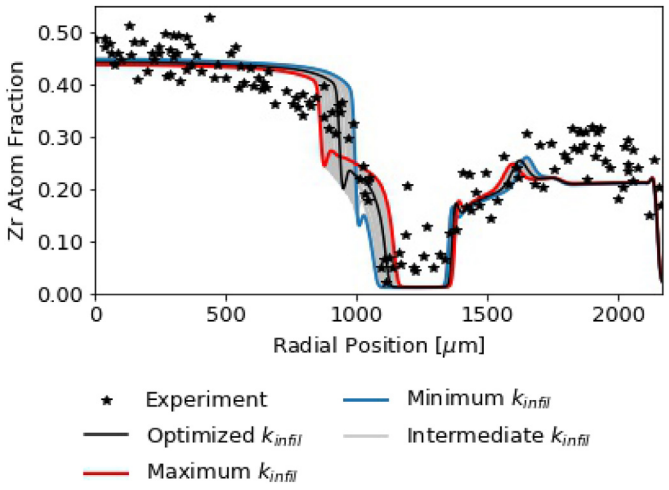


Fig. 14. Zr profiles predicted using different k_{infil} values. k_{infil} defines the fuel's steady-state thermal conductivity and is related to the fuel's total porosity and the amount of liquid sodium logged in the fuel. Variations in k_{infil} influence the steady-state temperature profile and kinetics, which impact the shape and location of the inner edge of the Zr bathtub.

variations can significantly alter constituent redistribution through the parameters' influences on fuel temperatures, phase boundary locations, and kinetics. Analysis of these sensitivities helps to build confidence in the understanding of the system and ability to predict its behavior.

The Zr profiles predicted using different q' values are shown in Fig. 15. The results show that variations in q' affect the shape of the inner edge of the Zr bathtub and the position of its outer edge. Higher values of q' raise the temperature of the fuel, enhancing kinetics and smoothing rough features in the Zr profiles. Fig. 8 shows that T_{CL} increases and phase boundaries shift outwards toward the fuel periphery, as expected.

Sensitivity analysis results for T_s are shown in Fig. 16. The results follow the same trends described above and show that variations in T_s drastically impact the model's predictions.

It is difficult to justify selecting one q' or T_s value over another just to improve model agreement, but the uncertainties in these parameters cannot be ignored. The model's predictions are best interpreted stochastically. We recommend careful characterization

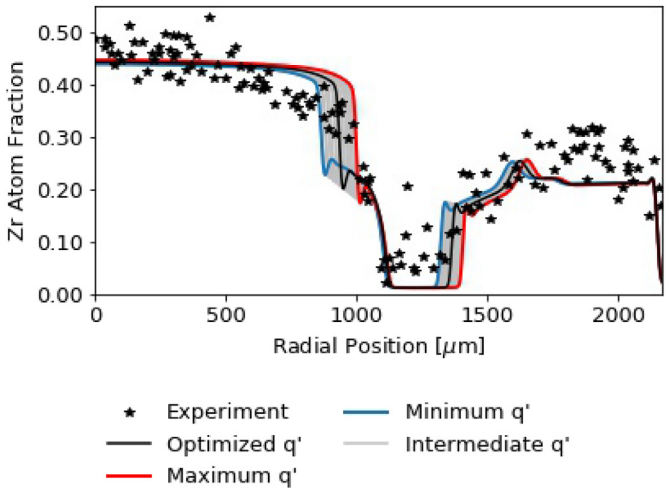


Fig. 15. Zr profiles predicted using different q' values. Variations in q' affect the shape and position of the Zr bathtub.

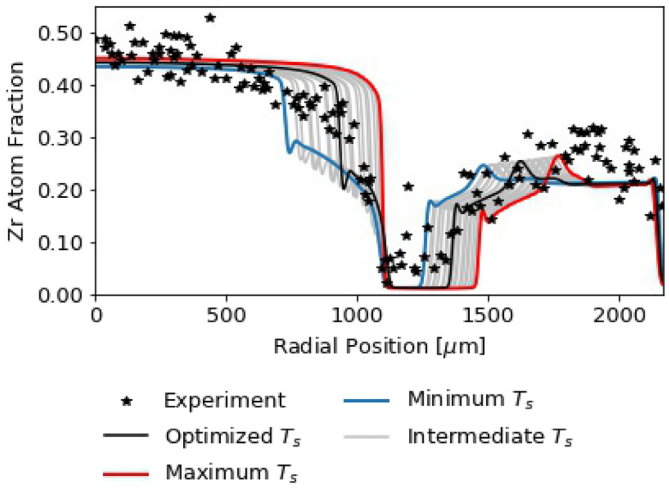


Fig. 16. Zr profiles predicted using twenty different T_s values.

of the uncertainties associated with each operational parameter so that the model's uncertainties can be quantified. Model predictions should then be presented with meaningful confidence levels, which would vastly improve the utility of fuel performance and safety calculations.

6. Conclusion

Constituent redistribution is a complex kinetic process driven by gradients in temperature and chemical potential. In U–Zr fuels, constituent redistribution promotes formation of radially-distributed phase fields, each with unique thermal, mechanical, and diffusive properties and behaviors. The phenomenon is closely coupled to swelling, fission gas release, and sodium infiltration behaviors through the fuel's temperature- and composition-dependent thermal conductivity. Despite numerous studies and modeling efforts, the ability to accurately model constituent redistribution is still limited, and serious questions about phase transition temperatures and irradiation-enhanced diffusion remain unanswered. These deficiencies must be addressed to model lifetime-limiting behaviors such as fuel-cladding chemical interaction and extrapolate beyond the existing experimental data for design and safety analyses.

All of the key metallic fuel behaviors must be accounted for to predict the coevolution of fuel temperature, phase, and constituent profiles with precision. Not all of these behaviors can be modeled mechanistically at this time due to lack of material property data and deficiencies in the understandings of each behavior and the interactions between. The behaviors must be investigated one or two at a time using an iterative approach to learn about the system and refine our predictive capabilities. In the current work, we developed a quantitative phase-field model of the U–Zr system using the most realistic thermodynamic and kinetic descriptions available. Heat conduction and transport components were incorporated to account for coupled fuel behaviors using suitable approximations. Model parameters were optimized using DP-81 PIE data and validated against PIE data from DP-11. Sensitivity analysis was then performed to investigate gaps in the knowledge of the system, identify deficiencies, and propose potential improvements.

We demonstrated that the model can predict constituent redistribution as accurately as previous models by optimizing only four parameters within physically reasonable bounds and without neglecting the effects of coupled irradiation behaviors. Agreement between model predictions and experimental data generally decreased with increasing burnup and was least favorable in fuel regions dominated by poorly characterized low-temperature phases. Increased uncertainties in swelling and sodium infiltration at high burnup and the overall lack of adequate material property data for the low-temperature phases likely contribute to these discrepancies. As in previous studies, diffusion needed to be increased to reproduce the observed behaviors. This observation is consistent with irradiation-enhanced diffusion, but further work is necessary to deconvolute these effects from those of parameter uncertainty. No modification of the phase transition temperatures was necessary in the current work, suggesting that the need for these modifications in some previous studies was due to parameter uncertainties or unrealistic simplifications or assumptions.

Sensitivity analysis results helped to characterize the roles of each model parameter and interpret the interactions between the various fuel behaviors. The majority of evolution occurs within the γ and $\beta+\gamma$ phase fields, and the mobile interface between the two regions is particularly influential in forming the Zr bathtub. Enhancement of β and γ kinetic parameters promotes outward and inward Zr migration, respectively. These parameters scale the overall magnitude of constituent redistribution while parameters associated with swelling and sodium infiltration influence the size, shape, and location of the Zr bathtub.

7. Recommendations

The current work brings the community closer to a comprehensive understanding of the irradiation behavior of metallic fuels and demonstrates that we have the tools necessary to begin to more mechanistically model these behaviors. The phase-field model is not intended to take the place of traditional macroscale fuel performance codes. Rather, it should be used to study the system's physics and refine the understanding of its material properties. These findings should then be incorporated into existing fuel performance codes.

Further experimental and modeling work is necessary to improve model performance and reliability. This will require data collection and refinement of the simplifications and approximations used to account for swelling, fission gas release, and sodium infiltration. These improvements will make it possible to extend models to the more complex U-Pu-Zr system and to investigate other proposed fuel geometries and compositions. We recommend that the following issues be addressed:

- Perform diffusion studies to better characterize the β kinetic parameters. Composition dependence may need to be taken into account.
- Study porosity development and evolution both before and after fission gas release. Characterization of the relationships between porosity and phase will be necessary to accurately model fuel elements at higher burnups.
- Determine how logged sodium content varies spatially in the fuel to increase the accuracy of temperature calculations.
- Incorporate the realistic thermodynamic and kinetic descriptions compiled for use in the current work into existing fuel performance codes.
- Account for coupling between irradiation behaviors when developing and refining metallic fuel performance models.
- Quantify and characterize the uncertainties associated with power and fuel surface temperature as more data is made available. Account for these uncertainties when modeling fuel behavior, and present meaningful confidence levels alongside model predictions.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study. The raw/processed data will be provided upon request.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jnucmat.2019.05.053.

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